# Skills Summary

## Interpreting Remainders in Real Contexts

*Use this reference while completing your worksheet or when studying.*

Every problem below starts from the same division, $74 \div 8 = 9$ R 2. The division never changes — the *question* decides which part of it is the answer.

### Skill 1 — Drop the remainder: complete groups only

**When:** the question asks how many *full*, complete or whole groups can be made.

**Answer:** the quotient. The leftovers cannot form a whole group, so they are not counted.

**Example:** 74 cookies are packed 8 to a bag. How many *full* bags?
**Step 1.** "Full bags" means only complete groups count, so the remainder is dropped and the quotient is the answer.
**Step 2.** $8 \times 9 = 72$, and $74 - 72 = 2$, so $74 \div 8 = 9$ R 2. The 2 extra cookies do not fill a bag: **9** full bags.

**Try it:** 59 pencils are packed 7 to a pack. How many full packs?

check: **8** full packs

### Skill 2 — Round up: nothing may be left behind

**When:** the question asks how many groups are *needed* so that every item (or person) is included.

**Answer:** the quotient + 1 whenever there is a remainder. The leftovers still need one more group.

**Example:** 74 cookies, 8 per box. How many boxes to hold *all* of them?
**Step 1.** "All of them" means the leftovers still need a container, so the quotient must be rounded up.
**Step 2.** $74 \div 8 = 9$ R 2. Nine boxes hold only 72 cookies, so the last 2 need a tenth box: **10** boxes.

**Try it:** 59 pencils, 7 per pack. How many packs to hold all of them?

check: **9** packs

## Skill 3 — The remainder is the answer

**When:** the question asks how many are *left over*, or how many are in the last partly filled group.

**Answer:** the remainder itself — found as remainder = total − (group size × quotient).

**Example:** 74 cookies, 8 per bag. How many are on the last, partly filled bag?
**Step 1.** The question asks about the leftovers only, so the answer is the remainder, not the quotient.
**Step 2.** $74 - (8 \times 9) = 74 - 72$, so the last bag holds **2** cookies.

**Try it:** 59 pencils, 7 per pack. How many pencils are in the last, partly filled pack?

check: **3** pencils

## Skill 4 — Estimate the quotient first

**When:** before dividing, to know roughly how big the answer should be (and to catch an answer that is ten times too big or too small).

**How:** round each number to the nearest ten, then divide the rounded numbers.

**Example:** 238 fans travel in buses that seat 40. Estimate the number of buses.
**Step 1.** An estimate only needs friendly numbers, so round first and divide the rounded numbers.
**Step 2.** 238 rounds to 240, and 40 stays 40, so $240 \div 40 =$ **6** buses — the exact answer should be near 6.

**Try it:** 179 chairs are set out in rows of 20. Estimate the number of rows.

check: **9** rows